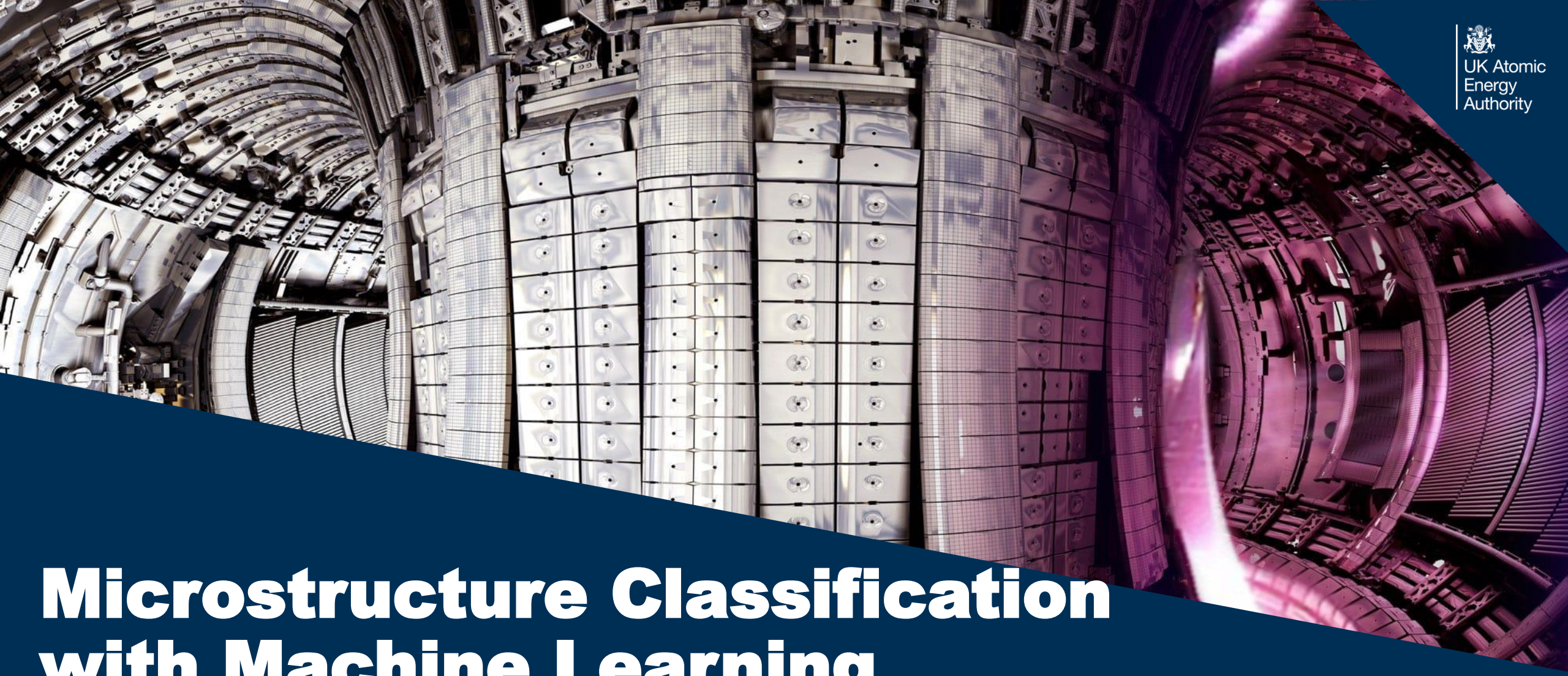


UK Atomic Energy Authority

Microstructure Classification with Machine Learning

Mike White

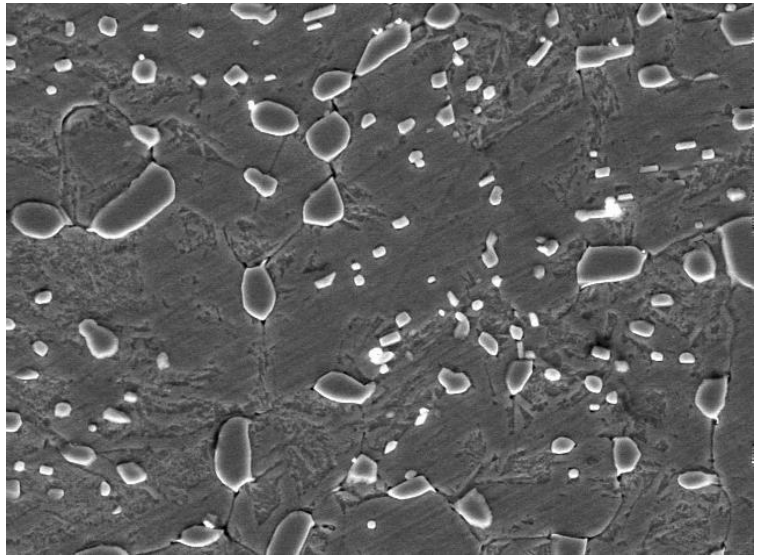
mike.white@ukaea.uk

16 March 2026

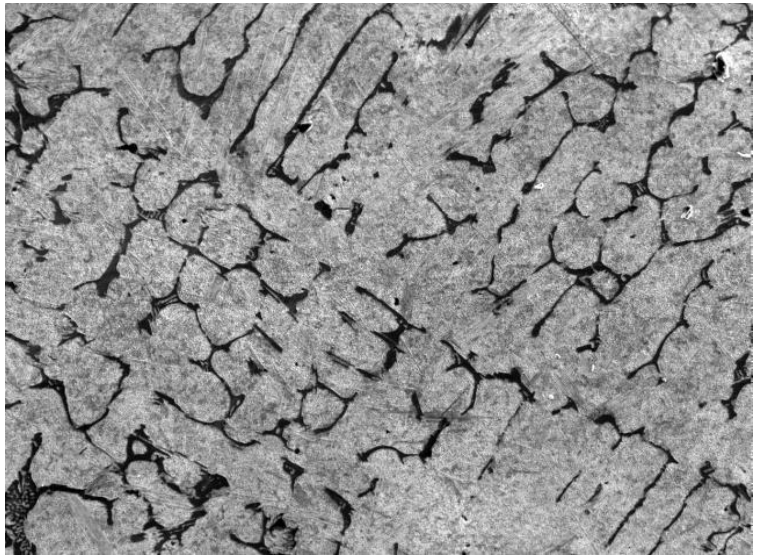
UKRI Engineering and Physical Sciences Research Council

MANCHESTER 1824 The University of Manchester

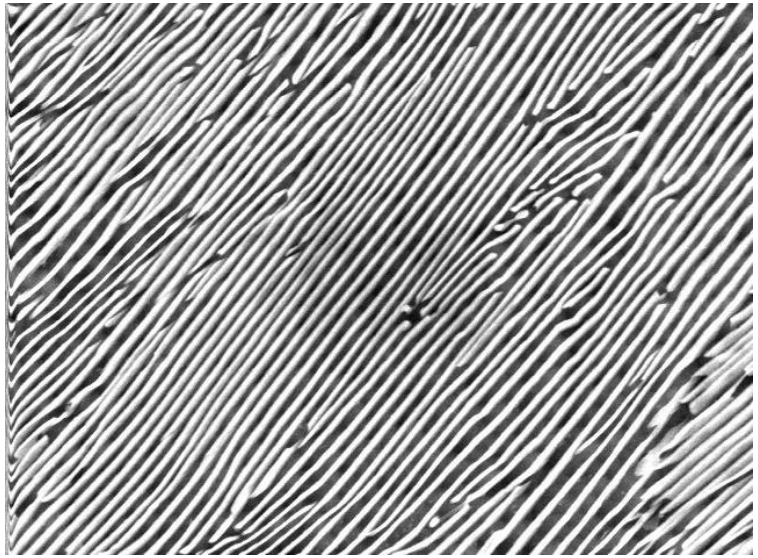
Why is microstructure important?



Spheroidite



Carbide Network



Pearlite

Process → Structure → Property

What is a “microstructural fingerprint?”

UK Atomic Energy Authority

- Compressed numerical representation of microstructure.
- Captures intrinsic properties of a given microstructure.
- Should be unique for a given microstructure.
- Applicable to a variety of data types.
- Application agnostic.
- Useful inputs for downstream tasks, such as:
  - Process-structure-property relationships,
  - Surrogate models,
  - Classification (quality control).

Why do we need them?

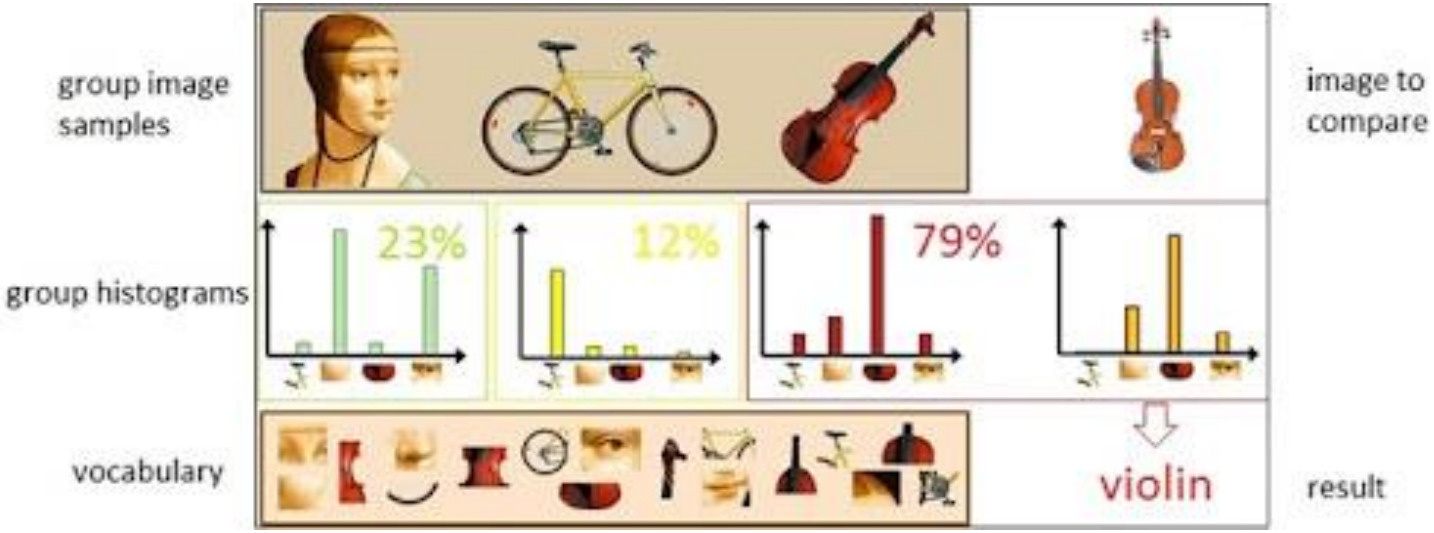
UK Atomic Energy Authority

- Microstructure can be thought of as a high-dimensional continuous function.
- Data collection discretises microstructure and generally captures a local snapshot of a global microstructure.
- Image capture yields high-dimensional data.
- Common methods for feature extraction (grain size, phase fraction, etc.) discard vast amounts of information.
- Suitable fingerprinting should capture the microstructure in general without the need to define specific features or metrics.
- Enables microstructure-dependent machine learning-based models.
- Today, we will compare two methods – local feature descriptors and convolutional neural networks (CNNs).

Local Descriptors

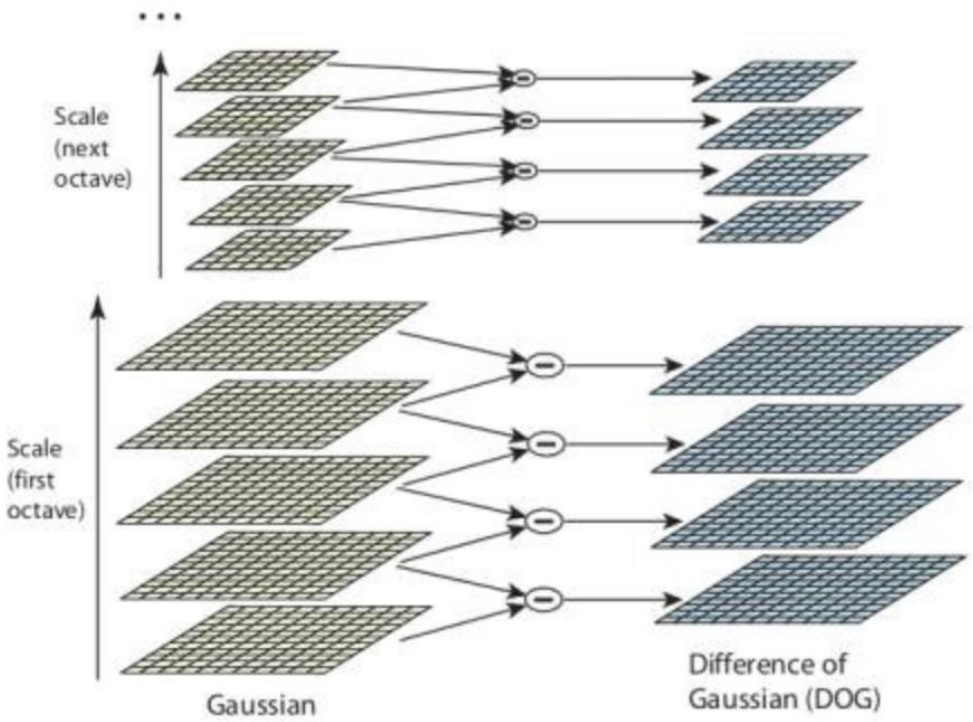
Overview

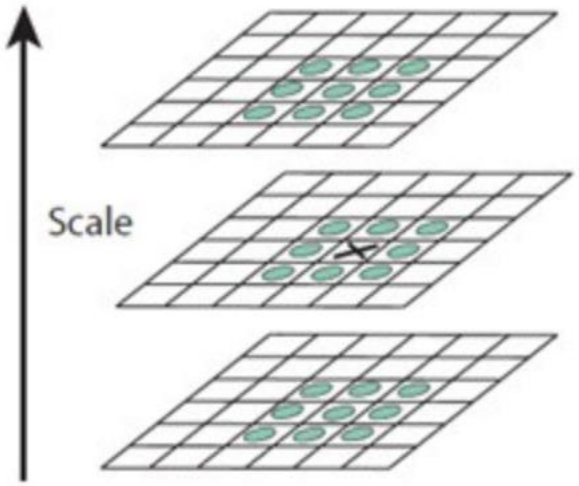
- First need to identify a set of keypoints where we are interested in describing the local behaviour in the image.
- Some region of pixels around the keypoint are then analysed to give a local description of the image at that keypoint.
- We then train a clustering algorithm to group similar features together.
- Once trained, we can then:
  - take a new image;
  - calculate the keypoint features;
  - apply the trained clustering algorithm to categorise the new features;
  - count the number of features in each category.



Keypoint-Based Features (Scale-Invariant Feature Transform)

- Keypoints identified with difference of Gaussians





Images from https://docs.opencv.org/3.4/da/df5/tutorial_py_sift_intro.html

Keypoint-Based Features (Scale-Invariant Feature Transform)

- Neighbourhood centred at keypoint described by histogram of orientation gradients.
- 8-bin histogram for 4x4 grid gives 128-dimensional feature vector for each keypoint.

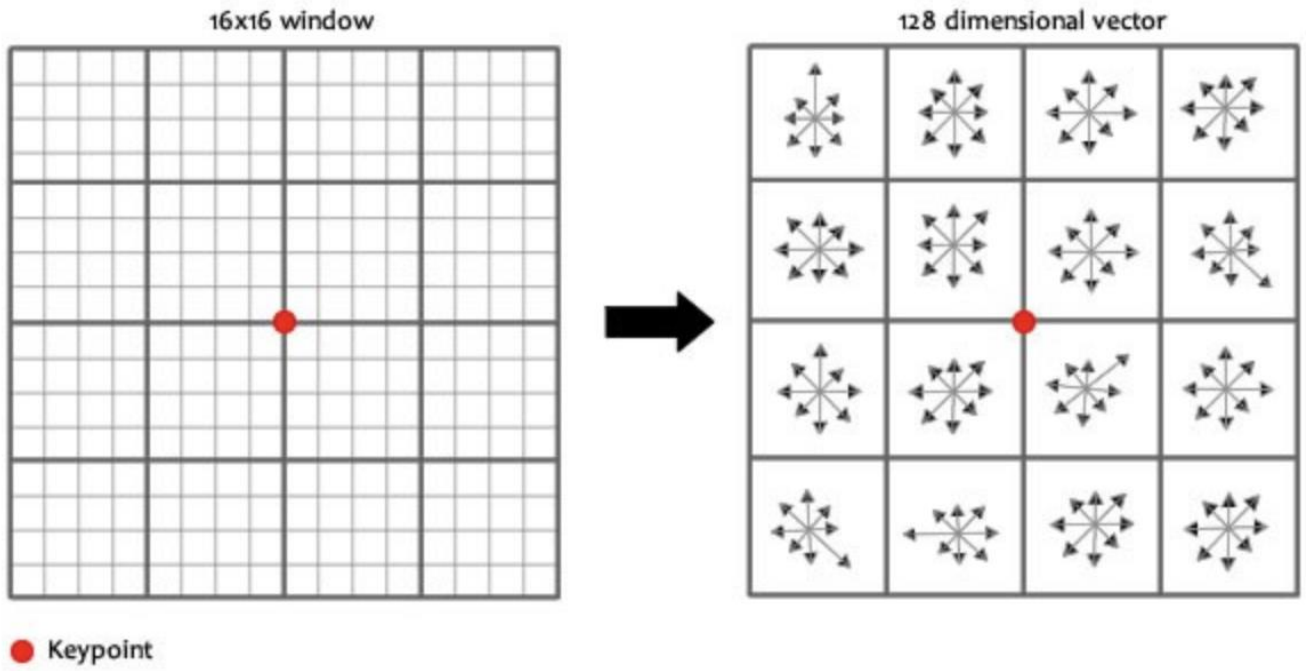
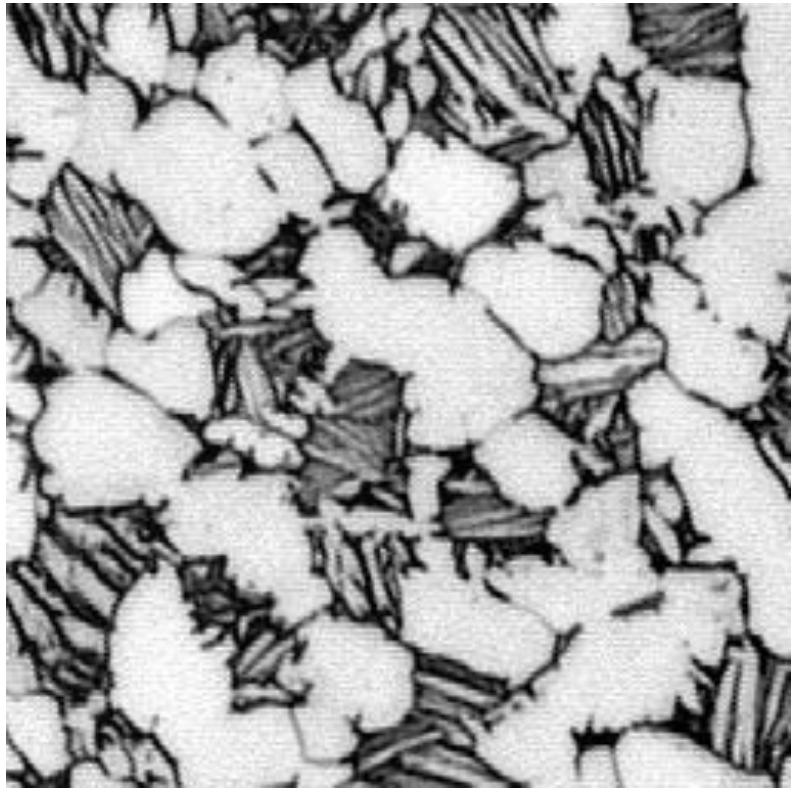
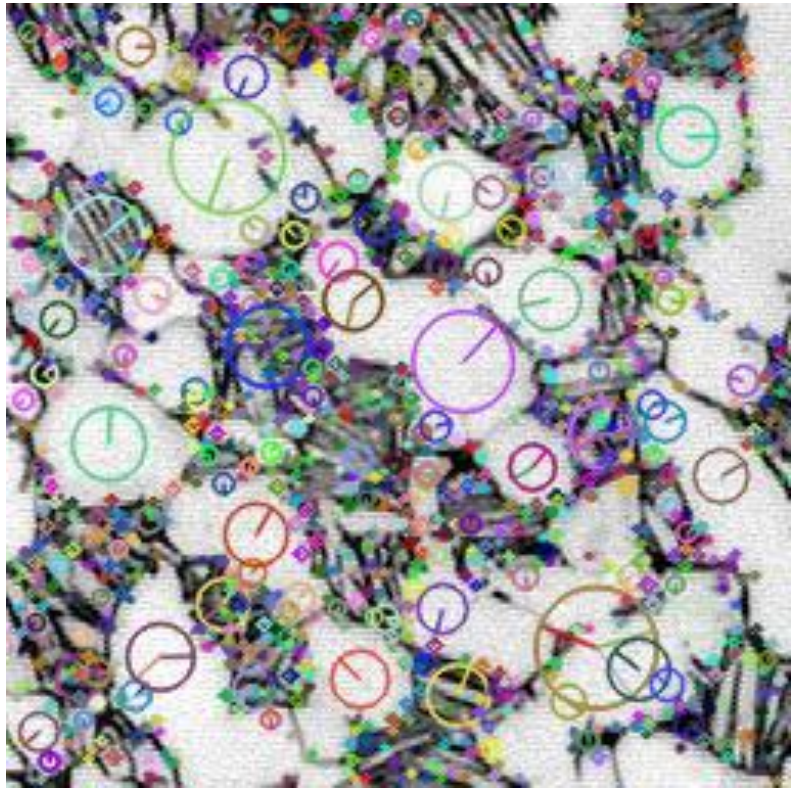


Image from https://medium.com/data-breach/introduction-to-sift-scale-invariant-feature-transform-65d7f3a72d40

Visual representation of SIFT

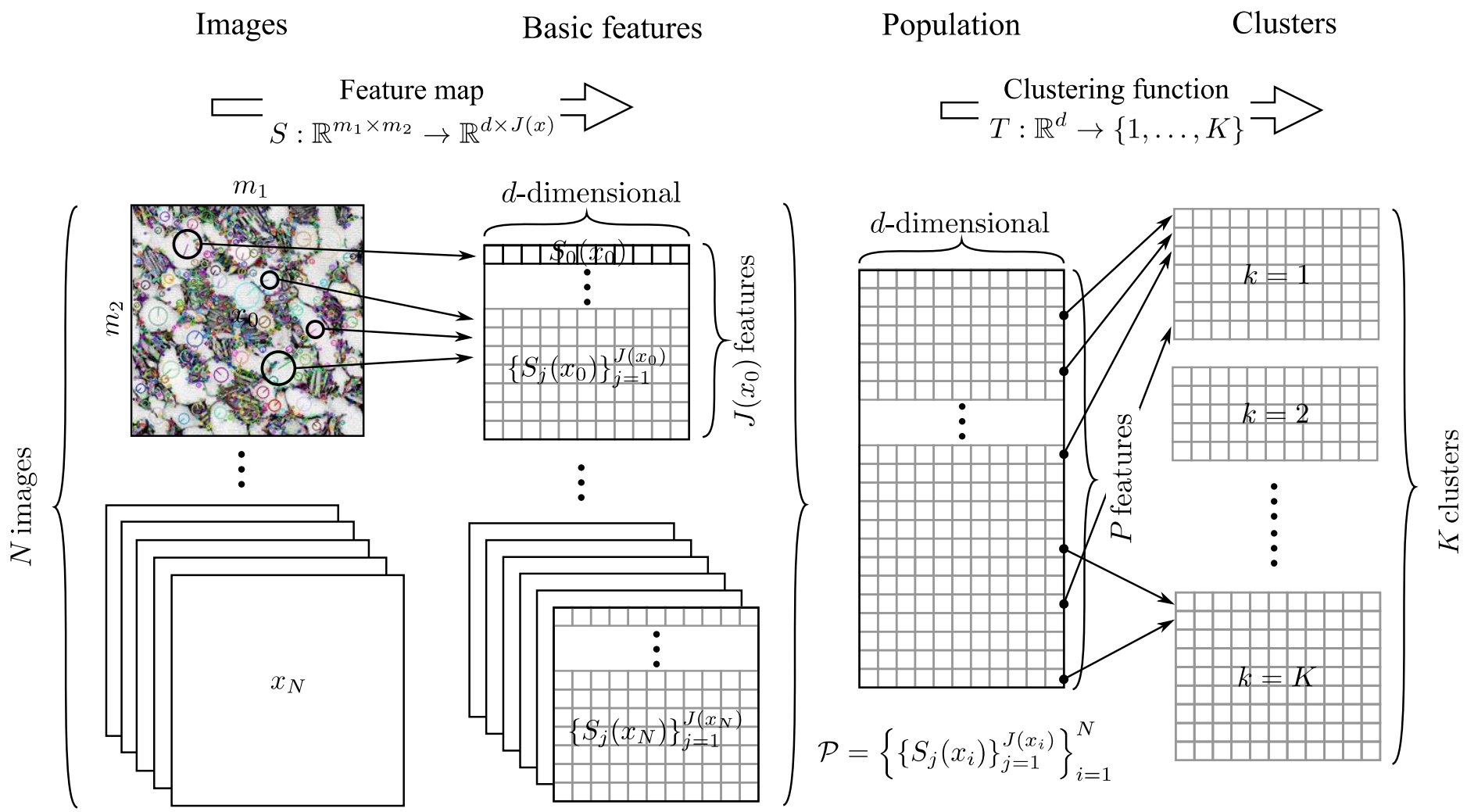


Bimodal Ti-6Al-4V optical micrograph

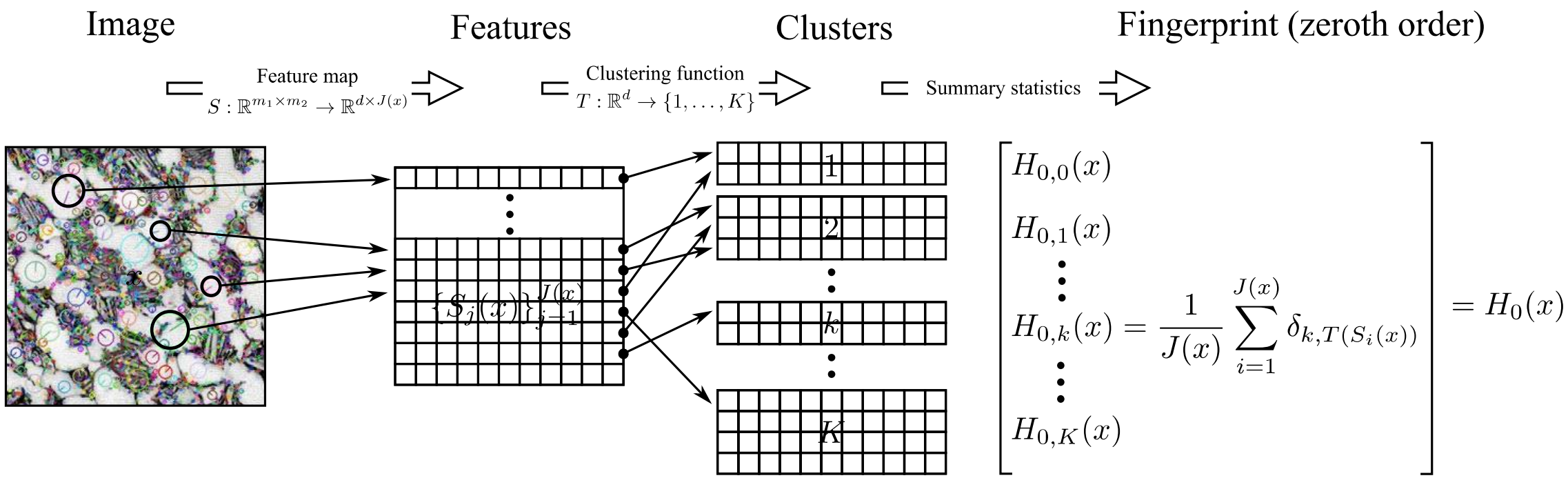


Visualisation of SIFT features

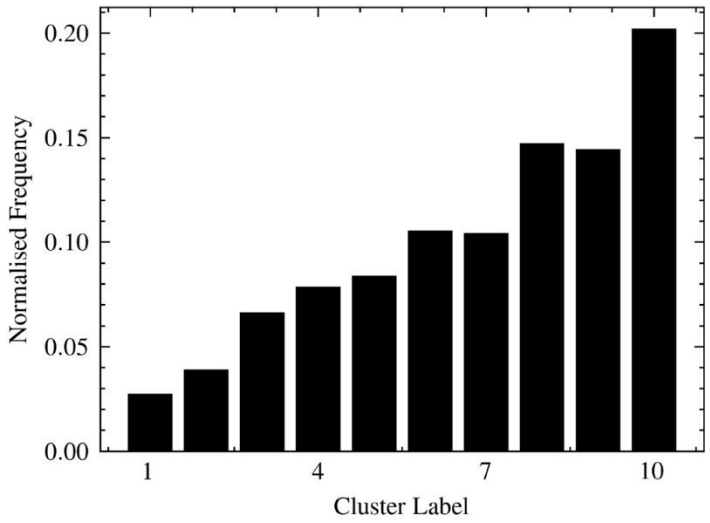
Statistical Framework (Training)



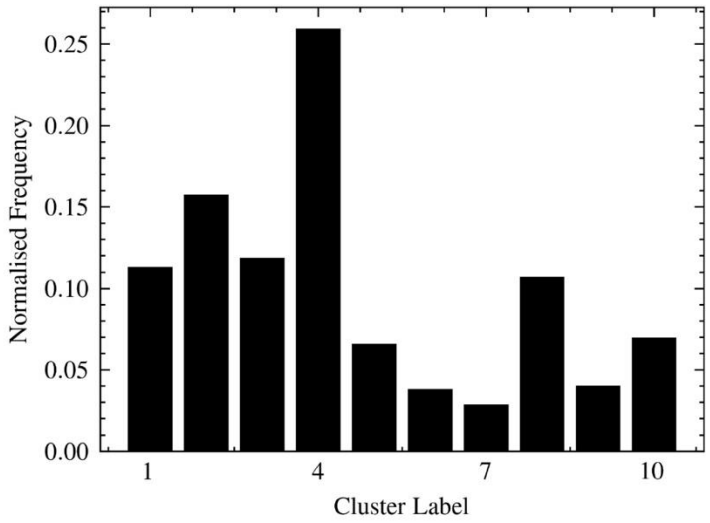
Statistical Framework (Fitting)



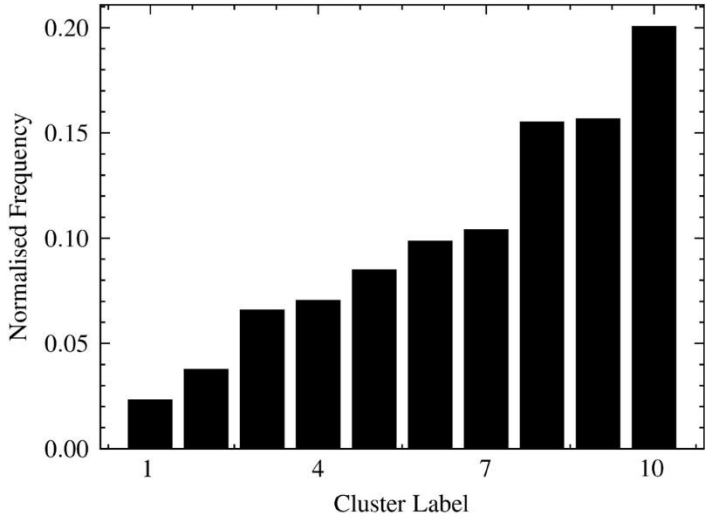
SIFT Fingerprints



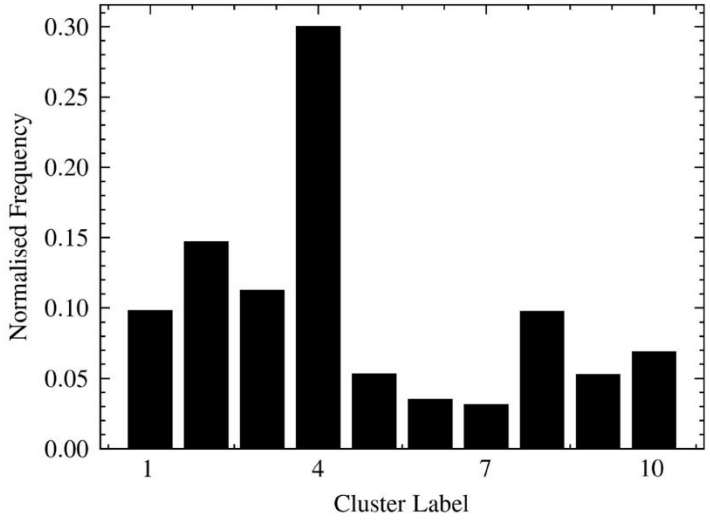
(a) Bi-modal



(b) Lamellar



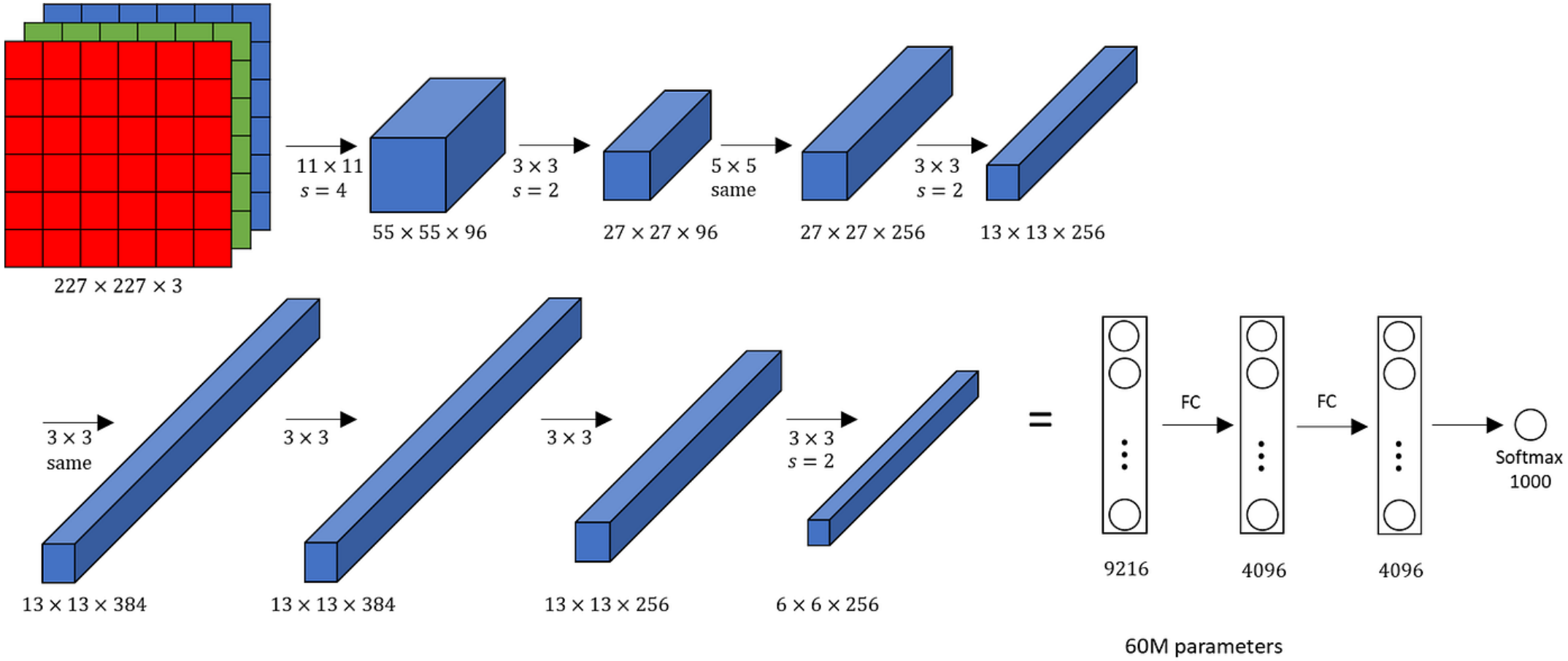
(c) Bi-modal



(d) Lamellar

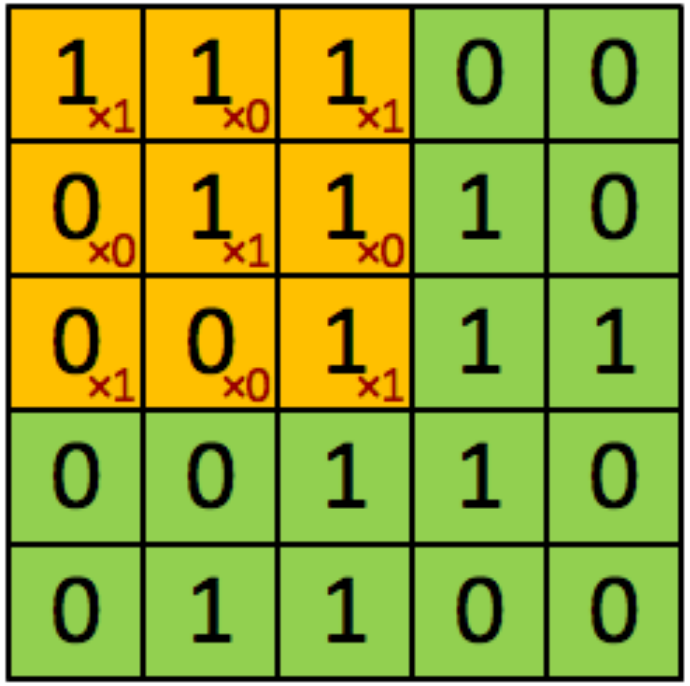
Convolution-Based Methods

Convolutional Neural Networks (CNNs) AlexNet

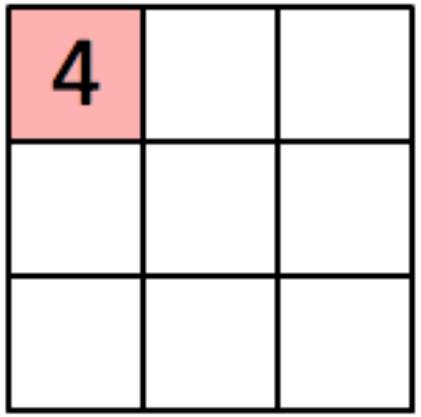


https://medium.com/@siddheshb008/alexnet-architecture-explained-b6240c528bd5

Filter Convolution



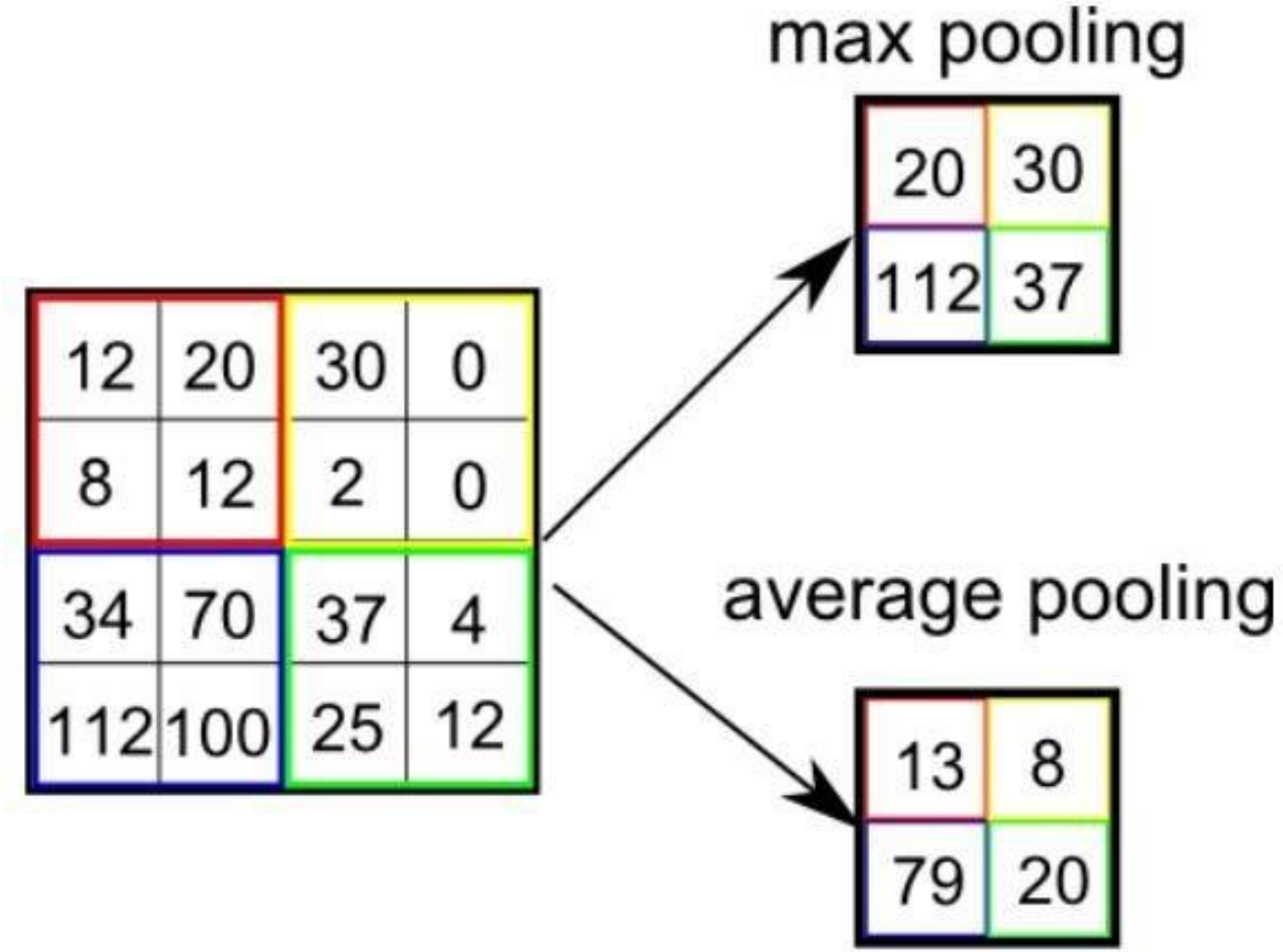
Image



Convolved Feature

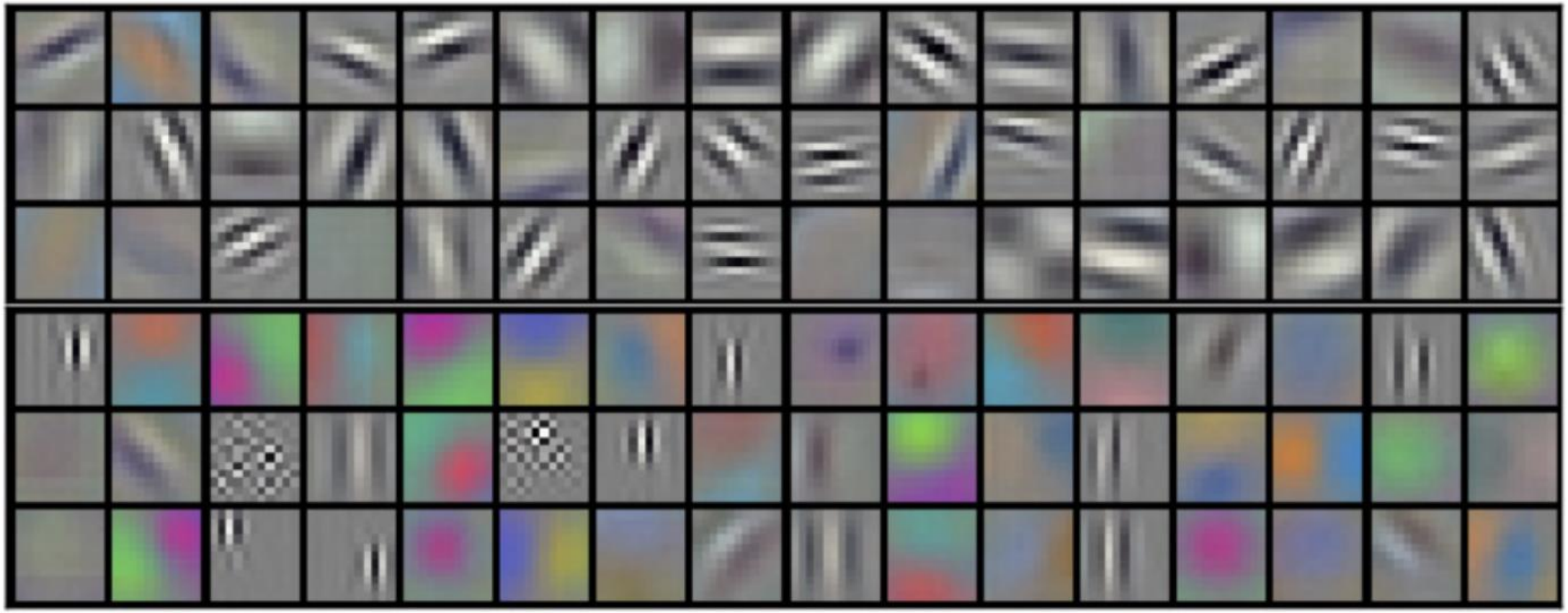
https://towardsdatascience.com/a-comprehensive-guide-to-convolutional-neural-networks-the-eli5-way-3bd2b1164a53

Pooling Layers



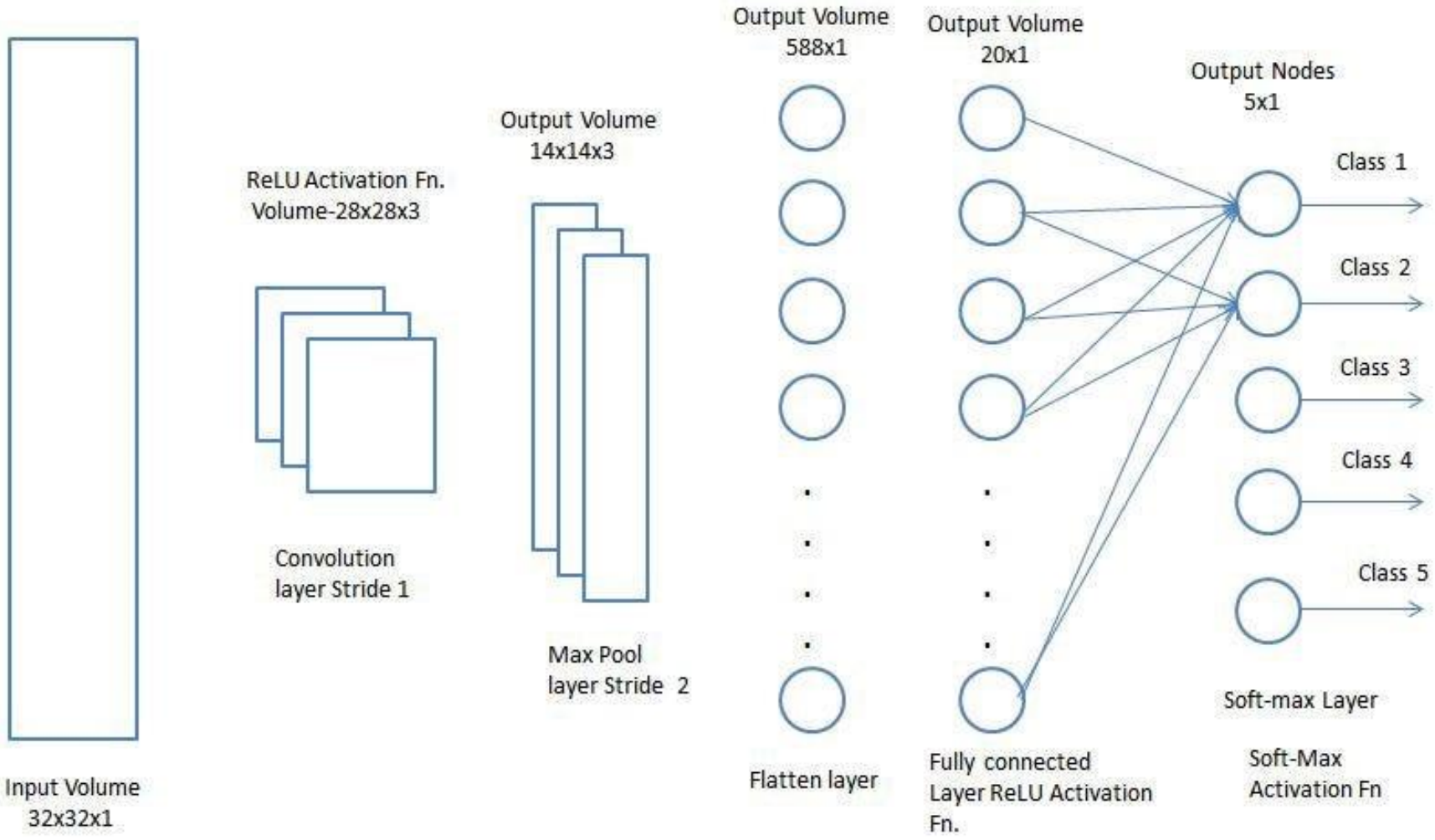
https://towardsdatascience.com/a-comprehensive-guide-to-convolutional-neural-networks-the-eli5-way-3bd2b1164a53

CNN Learned Filters



ImageNet Classification with Deep Convolutional Neural Networks, A. Krizhevsky, I Sutskever and G. Hinton, 2012

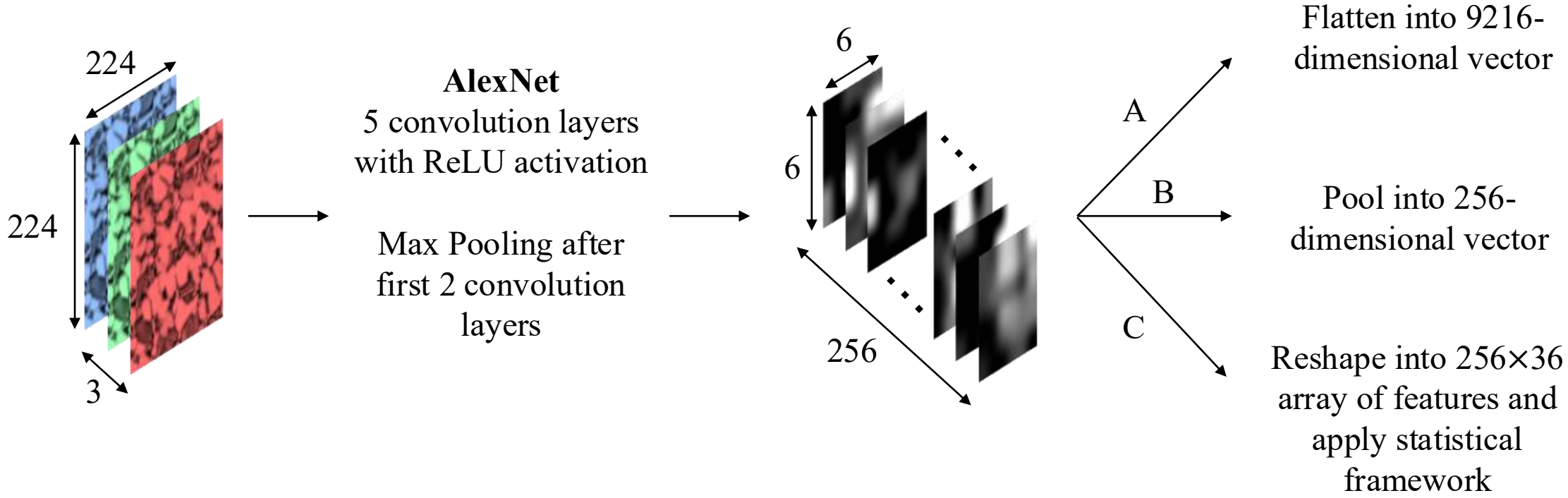
Classification



https://towardsdatascience.com/a-comprehensive-guide-to-convolutional-neural-networks-the-eli5-way-3bd2b1164a53

Transfer Learning

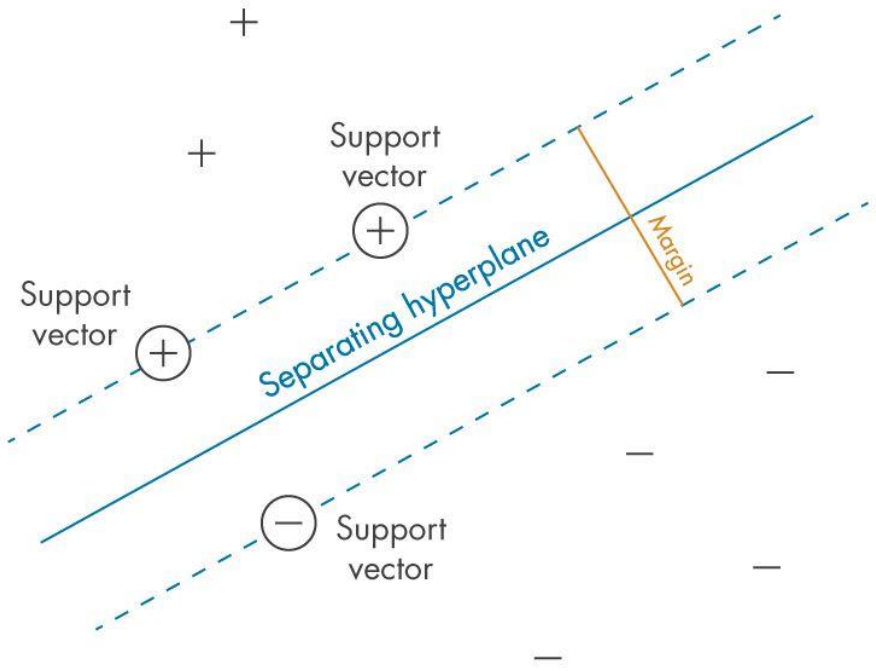
- Apply a pretrained model to new data.
- Pretraining not necessarily on microstructures.
- Fine tuning can be performed to optimise performance.
- When data is limited, fine tuning may not be required.



Support Vector Machine (SVM)

Classification

- We can now construct a fingerprint from an input image.
- These can now be used to train a classifier to learn a mapping between the fingerprint space to some set of class labels.
- Here, we will use a support vector machine (SVM).
- Say we have N classes that we are trying to learn.
- The main idea here is to construct a hyperplane in the fingerprint space that separates the space into N regions, where each region corresponds to a class label.
- Once trained, we can then input a fingerprint and see which region it belongs to.
- This then gives us the class label.



Objectives

UK Atomic Energy Authority

This was a brief overview of the methods we will be using in the workshop today.

We will now go through the provided notebook to see how these methods are implemented in Python.

The main aims of this workshop are:

- To understand some machine learning models that are available for image classification.
- Compare the performance of a more traditional method (keypoint features) against a more modern approach (CNNs).

The notebook for todays workshop is available at:

https://github.com/mikedwhite/MicrostructureClassificationWorkshop

Please navigate there now and follow the instructions for getting started in Google Colab.